
LECTURE NOTES

Lecture 13: Constrained Optimization

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1 Constrained Optimization

Minimizing an objective subject to design point restrictions called **constraints**

- New optimization problem statement

$$\begin{aligned} \min_x f(x) \\ \text{s.t. } x \in \mathcal{X} \end{aligned}$$

Where $\mathcal{X}$ is the **feasible set**

- $\mathbb{R}^n$ or $\mathbb{Z}^m$ constrained to a subset

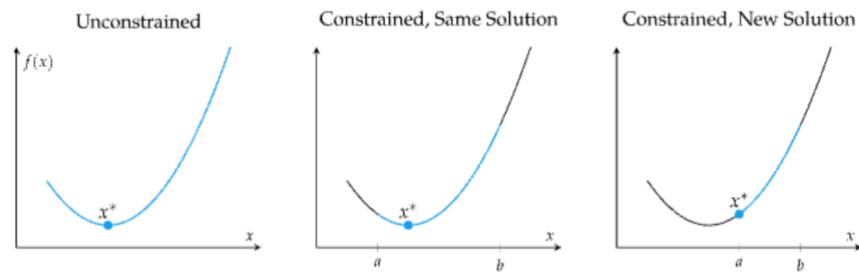


Figure 1: 1 dimension of constraints, where the solution changes

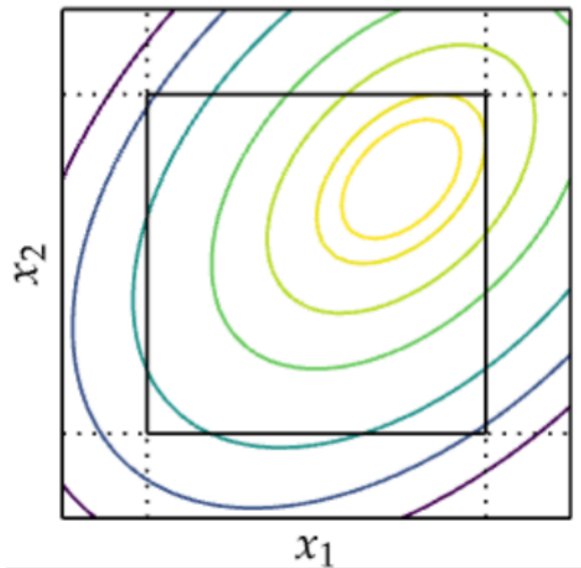


Figure 2: 2 dimensions

- Generally, constraints are formulated using two types:
 - (a) Equality constraints: $h(x) = 0$
 - (b) Inequality constraints $g(x) \leq 0$

2 Transformations to Remove Constraints

If necessary, some problems can be reformulated to use constraints in the objective function

Let x be constrained between a and b

$$x = t_{a,b}(\hat{x}) = \frac{b+a}{2} + \frac{b-a}{2} \left(\frac{2\hat{x}}{1+\hat{x}^2} \right)$$

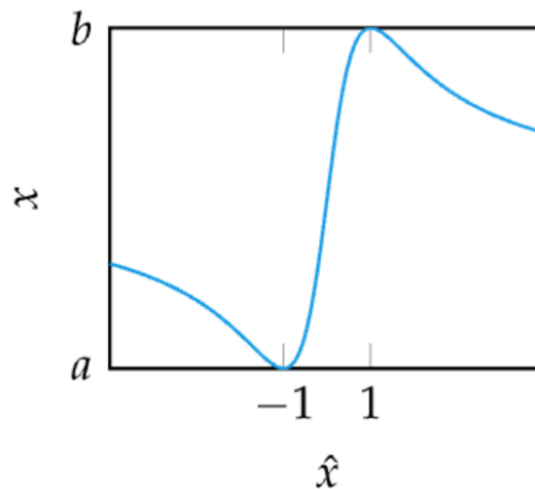


Figure 3: transformation

3 Lagrangian Multiplier Method

The method of Lagrangian Multipliers is used to optimize a function subject to (equality) constraints

(only equality constraints, critical points, because gradient of f and the gradient of h are aligned)

$$\min_x f(x)$$

So instead of $\text{s.t. } x \in \mathcal{X}$

We can use the **Lagrangian function**

- $\mathcal{L}(x, \lambda) = f(x) - \lambda h(x)$
- Then $\nabla_x \mathcal{L}(x, \lambda) = 0$ and $\nabla_\lambda \mathcal{L}(x, \lambda) = 0$
- This gets $\nabla f(x) = \lambda \nabla h(x)$ and $h(x) = 0$
- Then solve for x and λ

Intuitively, the method of Lagrange multipliers finds the point x^* where the constraint function is orthogonal to the gradient (since ∇h is normal to h)

Note: this only works with equality constraints

3.1 Inequality Constraints

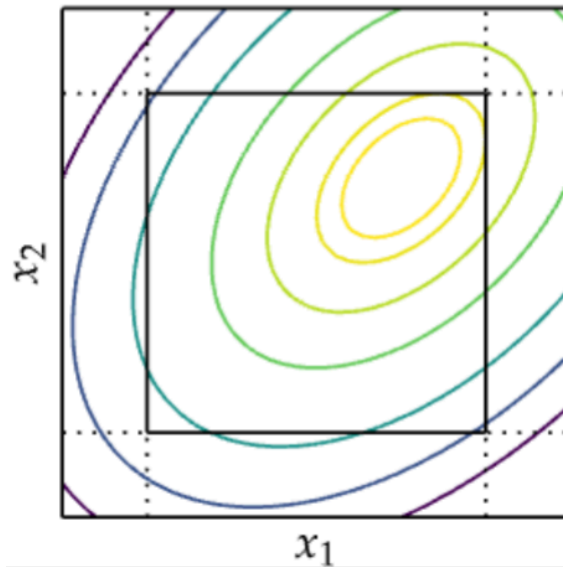


Figure 4: For inequality constraints, the local behavior for the gradient in the exact meeting point of two constraint functions is a ned gradient that cannot be improved.

$$\nabla f(\vec{x}^*) = \lambda_1 \nabla g_1(\vec{x}^*) + \lambda_2 \nabla g_2(\vec{x}^*)$$
$$\lambda_1, \lambda_2 \geq 0$$

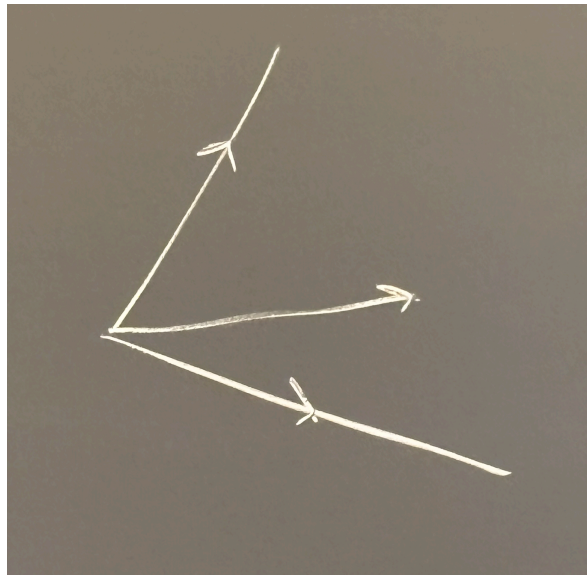


Figure 5: The new gradient vector composed of two gradients

Lets do the Langrangian Function Method

$$\mathcal{L}_\infty(x) = f(x) + \infty(g(x) > 0)$$

Though this is impractical because it is discontinuous and nondifferentiable.

- Instead, for $\mu \geq 0$:
 - (a) $\mathcal{L}(x, \mu) = f(x) + \mu g(x)$
 - (b) $\mathcal{L}_\infty(x) = \max_{\mu} \mathcal{L}(x, \mu)$
- For x infeasible $\mathcal{L}_\infty(x) = \infty$; for x feasible $\mathcal{L}_\infty(x) = f(x)$
- The new optimization problem becomes

$$\min_x \max_{\mu \geq 0} \mathcal{L}(x, \mu)$$

This is called the **primal problem**

3.2 Necessary Conditions - KKT Conditions (Karush-Kuhn-Tucker)

For (FONC) x^* to be a critical point then we need

$$\begin{cases} \nabla f(\bar{x}^*) = \bar{\lambda} \nabla \bar{g}(\bar{x}^*) \\ \bar{g}(\bar{x}^*) \geq 0 \\ \bar{\lambda} \geq 0 \\ \bar{\lambda} \cdot \bar{g}(\bar{x}^*) = 0 \end{cases}$$

Particular cases:

- f is concave, g is convex, then KKT are also sufficient
- x^* interior point
- Patological cases where they do not hold (formal expressions of the conditions include assumptions to avoid these cases).
- Generalized Lagrangian Function:

$$\mathcal{L}(x, \mu, \lambda) = f(x) + \sum_i \mu_i g_i(x) + \sum_i \lambda_i h_i(x)$$

- The **primal form** of the optimization problem

$$\min_x \max_{\mu \geq 0, \lambda} \mathcal{L}(x, \mu, \lambda)$$

- Reversing the order of operations leads to the **dual form**

$$\max_{\mu \geq 0, \lambda} \min_x \mathcal{L}(x, \mu, \lambda)$$

Theorem (Max-min inequality)

For any function $f : Z \times W \rightarrow \mathbb{R}$,

$$\underbrace{\sup_{z \in Z} \inf_{w \in W} f(z, w)}_{\text{dual}} \leq \underbrace{\inf_{w \in W} \sup_{z \in Z} f(z, w)}_{\text{primal}}$$

For us:

$$\max_{\mu \geq 0, \lambda} \min_x \mathcal{L}(x, \mu, \lambda) \leq \min_x \max_{\mu \geq 0, \lambda} \mathcal{L}(x, \mu, \lambda)$$

Therefore, the solution to the dual problem d^* is a lower bound to the primal solution p^*

4 Penalty methods

Example

$$\begin{array}{ll} \underset{\mathbf{x}}{\text{minimize}} & f(\mathbf{x}) \\ \text{subject to} & \mathbf{g}(\mathbf{x}) \leq \mathbf{0} \\ & \mathbf{h}(\mathbf{x}) = \mathbf{0} \end{array} \qquad \begin{array}{ll} \underset{\mathbf{x}}{\min} & f(\mathbf{x}) + \rho \cdot p_{\text{count}}(\mathbf{x}) \\ \text{s.t.} & p_{\text{count}}(\mathbf{x}) = \sum_i (g_i(\mathbf{x}) > 0) + \sum_j (h_j(\mathbf{x}) \neq 0) \end{array}$$

Figure 6: Penalty by counting violations

Procedure penalty_method;

Input: $f, p, \mathbf{x}, k_{\max}; \rho = 1, \gamma = 2$

Output: $\mathbf{x}$ solution

for k in $1, \dots, k_{\max}$ **do**

$\mathbf{x} \leftarrow \underset{\mathbf{x}}{\text{minimize}} \{f(\mathbf{x}) + \rho \cdot p(\mathbf{x})\};$

$\rho \leftarrow \rho \cdot \gamma;$

if $p(\mathbf{x}) = 0$ **then**

return $\mathbf{x};$

return $\mathbf{x};$

Figure 7: penalty methods

- Count penalty:

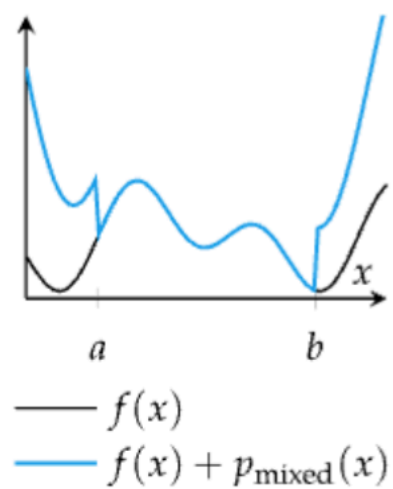
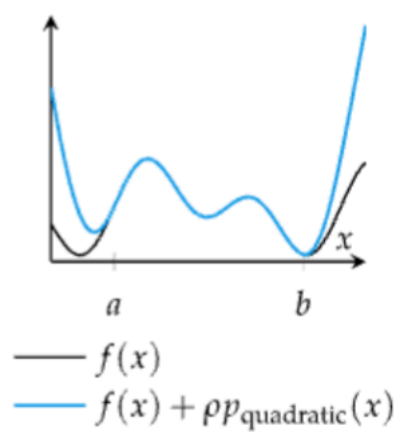
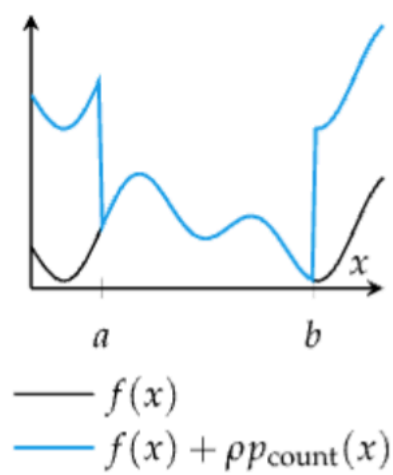
$$p_{\text{count}}(x) = \left(\sum_i g_i(x) > 0 \right) + \sum_j (h_j(x) \neq 0)$$

- Quadratic penalty:

$$p_{\text{quadratic}}(x) = \sum_i \max(g_i(x), 0)^2 + \sum_j h_j(x)^2$$

- Mixed penalty:

$$p_{\text{mixed}}(x) = \rho_1 p_{\text{count}}(x) + \rho_2 p_{\text{quadratic}}(x)$$



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Figure 8: multiple penalty methods

4.1 Augmented Lagrange Method

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Procedure augmented_lagrange_method;  
Input:  $f, h, \mathbf{x}, k_{max}; \rho = 1, \gamma = 2)$   
 $\lambda \leftarrow \mathbf{0};$   
for  $k$  in  $1, \dots, k_{max}$  do  
     $p \leftarrow (x \mapsto \rho/2 \cdot \sum_i (h_i(\mathbf{x}))^2) - \lambda \cdot h(\mathbf{x});$   
     $\mathbf{x} \leftarrow \text{minimize}_{\mathbf{x}} \{f(\mathbf{x}) + p(\mathbf{x})\};$   
     $\lambda \leftarrow \lambda - \rho \cdot h(\mathbf{x});$   
     $\rho \leftarrow \rho \cdot \gamma;$   
return  $\mathbf{x};$ 
```

Figure 9: Adaptation of penalty method for equality constraints

$$p_{\text{Lagrangian}}(x) \stackrel{\text{def}}{=} \frac{1}{2} \rho \sum_i (h_i(x))^2 - \sum_i \lambda h_i(x)$$

4.2 Interior Point Methods

Also called **barrier methods**, interior point methods ensure that each step is feasible

This allows premature termination to return a nearly optimal, feasible point

Barrier functions are implemented similar to penalties but must meet the following conditions:

- (a) Continuous
- (b) Non-negative
- (c) Approach infinity as x approaches boundary

Methods

- Inverse Barrier:

$$p_{\text{barrier}}(x) = - \sum_i \frac{1}{g_i(x)}$$

- Log Barrier

$$p_{\text{barrier}}(x) = - \sum_i \begin{cases} \log(-g_i(x)) & \text{if } g_i(x) \leq -1 \\ 0 & \text{if otherwise} \end{cases}$$